Maintenance Project Phase 1

1- GENERAL REQUIREMENTS

1.1- System Architecture

- The system is implemented as a **RESTful web service** using **Java Spring Boot**. It exposes endpoints grouped by functionalities (authentication, courses, lessons, assessments, users, etc.).
- All data is temporarily stored using **in-memory static maps**, simulating persistent storage. This setup is suitable for prototyping or early-phase testing.
- The system supports three distinct roles:
    - Admin
    - Instructor
    - Student

1.2- Authentication and Security

- Users log in using their email and password.
- Role-based access control ensures users only access operations relevant to their role.

1.3- Notifications System

- Every user can receive and manage notifications.
- Notifications are triggered by system events like course enrollment, course assignment, and content updates.
- Users can filter their notifications (e.g., unread, all) and mark notifications as "read" for tracking purposes.

2- USER MANAGEMENT

2.1- Admin Role

Admins act as super-users and system managers. Their core functionalities include:

- **User Management**:
    - Create new users and assign roles (Admin, Instructor, Student).
    - View a list of all registered users.
    - Filter user listings based on roles.
- **Course Management**:
    - Create new courses by providing:

      -Title
      - Description
      -Duration
      -Media files (e.g., videos, PDFs).
    - Edit course details including the list of lessons.
    - Delete courses and notify impacted users.
- **Monitoring and Oversight**:
    - Monitor system usage by role distribution.
    - Ensure instructors are properly assigned and students are enrolled.

2.2- Instructor Role

Instructors are course content creators and evaluators. Their capabilities include:

- **Course Participation**:

    - Can be assigned to one or multiple courses by Admins.
    - Cannot create or delete courses independently.

- **Lesson & Attendance**:
    - Add and update lessons within assigned courses.
    - Generate unique OTPs (One-Time Passwords) for each lesson to be used by students for attendance.
    - View attendance records per lesson.

- **Assessment Management**:
    - Create quizzes by selecting questions from a shared question bank.
    - Create assignments with descriptive instructions.
    - Add new questions to the question bank.
    - Grade quiz and assignment submissions and assign marks and feedback.

- **Student Performance Tracking**:
    - View aggregated performance for all students in their courses.
    - Filter performance by:

        -All Grades

        -Assignment Grades

        -Quiz Grades

2.3- Student Role

Students are the consumers of learning content. Their interactions include:

- **Course Interaction**:
    - Browse available courses.
    - Enroll in selected courses.

- **Attendance & Lessons**:
    - View lesson content.
    - Mark attendance using the OTP shared by the instructor.
- **Assessment Submission**:
    - Take quizzes created by instructors.
    - Submit assignments, including a file name or content upload.
- **Performance Monitoring**:
    - View personal grades, feedback, and progress in enrolled courses.
    - Filter views by quizzes or assignments.

3- COURSE MANAGEMENT

3.1- Course Entity

Each course includes:

- A unique identifier (ID)
- Title
- Description
- Duration (in hours, weeks, or a suitable unit)
- Media files (could be links to videos, PDFs, etc.)
- An assigned instructor
- A list of enrolled students
- A list of associated lessons

Admin Functions

- Create, update, and delete courses.
- Assign an instructor to a course.
- Update media and lesson content within a course.

Instructor Functions

- View assigned courses.
- Create and manage lessons for those courses.

Student Functions

- View a list of available courses.
- Enroll in selected courses.

3.2- Lesson Entity

Each lesson within a course contains:

- ID
- Title
- Text content
- An OTP code (generated by the instructor)
- Attendance map per student (tracks attendance using the OTP system)

Lessons are used to:

- Deliver course content.
- Track which students attended each session using OTPs.

4- ASSESSMENT MANAGEMENT

4.1- Quiz Entity

Quizzes are used for testing students and are defined as follows:

- Created by instructors.
- Consist of:
    - Title
    - Total marks
    - Number of questions
- Questions are pulled from a shared question bank.
- Upon quiz submission:
    - System automatically evaluates answers.
    - Calculates the number of correct responses.
    - Provides feedback and stores grading records.

Student Functions:

- Take assigned quizzes.
- View quiz scores and feedback post-submission.

Assignment Entity

Assignments allow deeper evaluation and creativity. They include:

- Created by instructors.

- Defined with a title and description.
- Students upload responses by attaching a file or providing a file name.
- Instructors assign grades and provide feedback.

The system stores:

- Submitted files
- Marks assigned
- Instructor feedback

5- NOTIFICATIONS

The notification module automatically informs users about important system events.

5.1- System-generated Notifications:

- When a student enrolls in a course:
    - Notification sent to the student and the assigned instructor.
- When an instructor is assigned to a course:
    - Instructor receives a notification.
- When a course is updated or deleted:
    - All enrolled students and the instructor are notified.

5.2- User Actions:

- View all notifications.
- Filter to view only unread ones.

- Mark notifications as read.

6- PERFORMANCE TRACKING

Instructors have access to student analytics across all their courses.

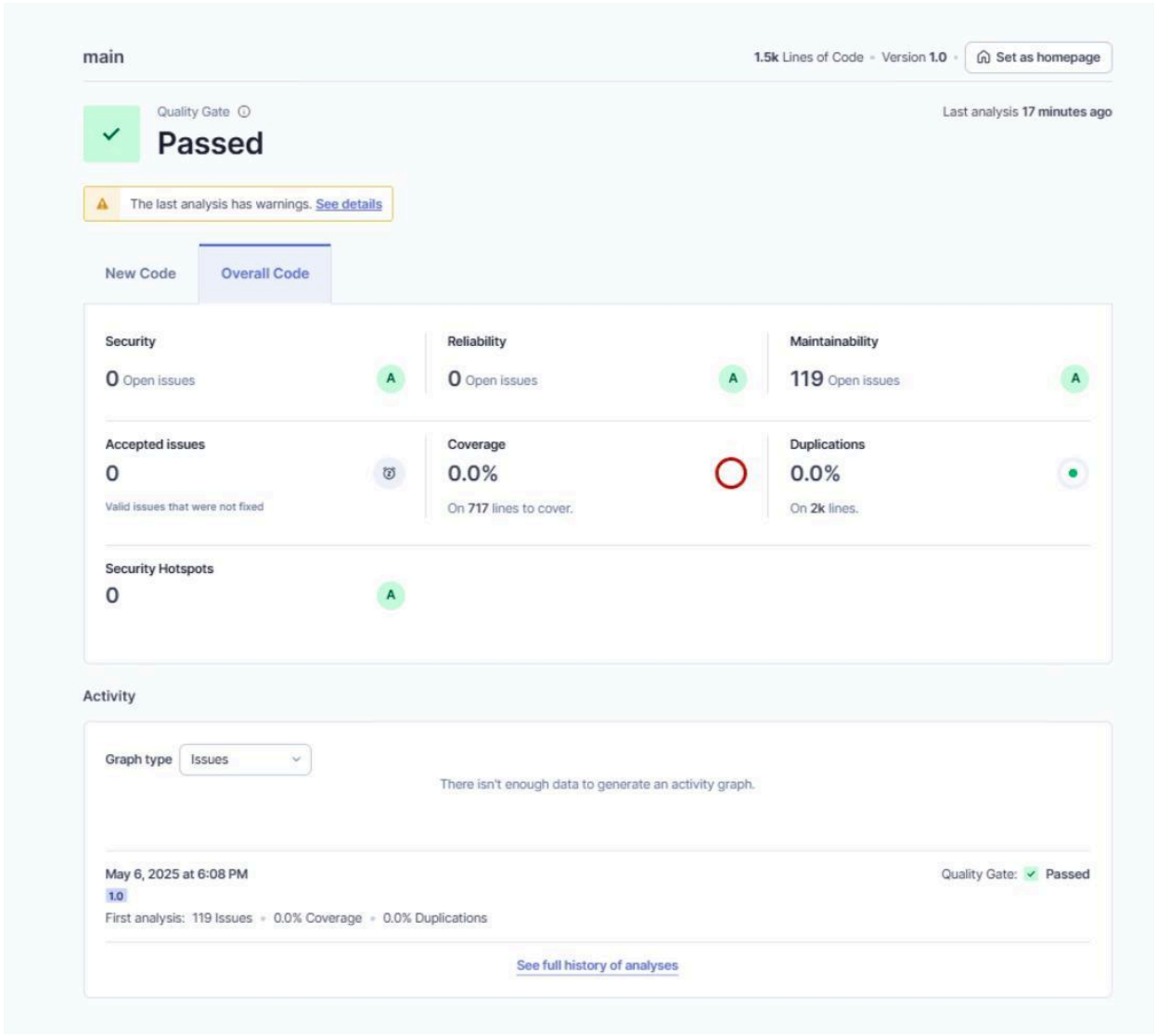
6.1- Available Metrics:

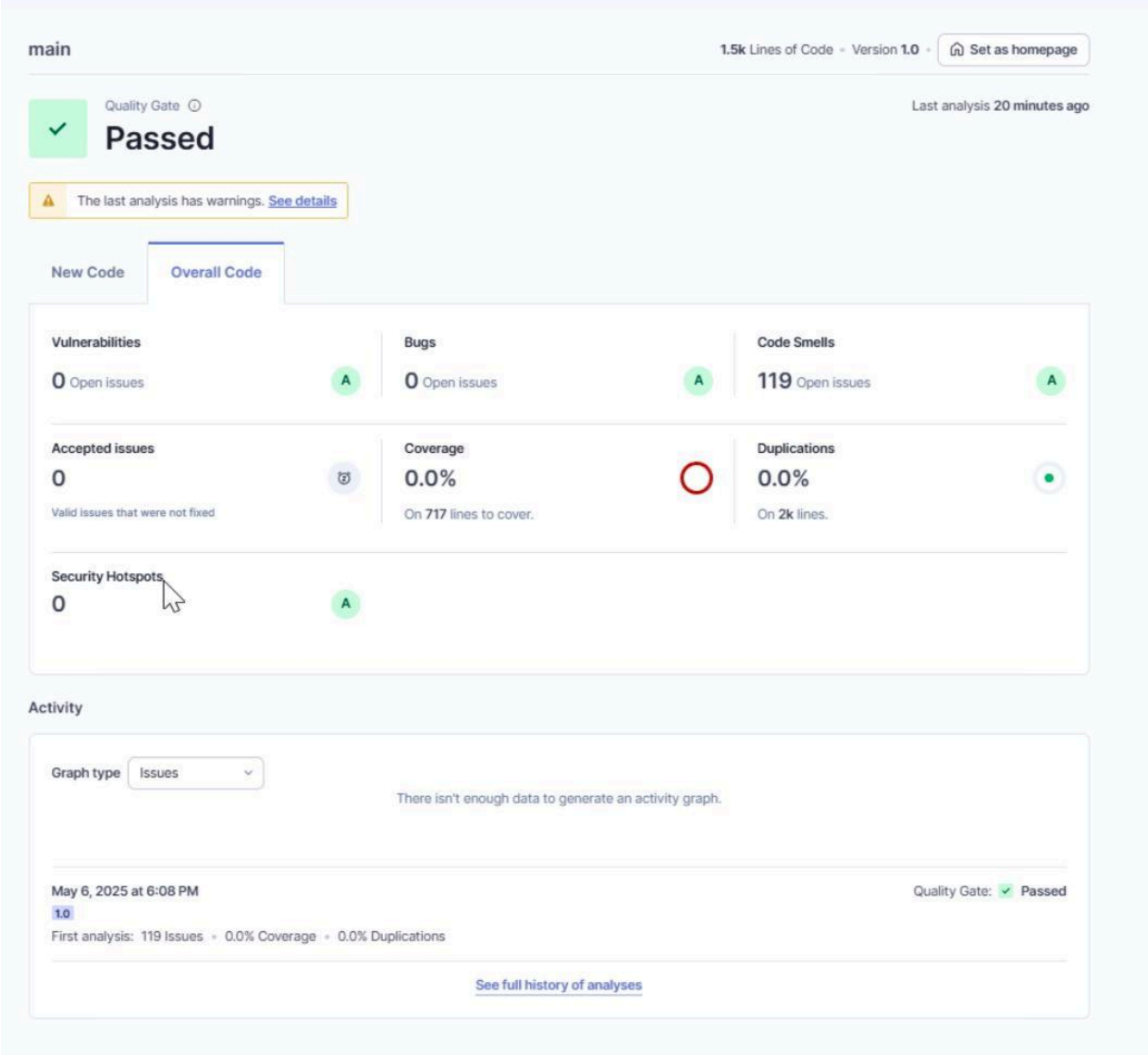
- Overall performance summary for each student.
- Quiz-specific results, including:
    - Number of attempts
    - Scores per attempt
- Assignment-specific results, including:
    - Submission status
    - Marks and feedback

Performance data helps instructors:

- Identify at-risk students.
- Adjust teaching or assessment strategies1.

7- The analysis report


main
1.5k Lines of Code • Version 1.0 • Set as homepage
Quality Gate
Passed
Last analysis 17 minutes ago
The last analysis has warnings. See details
New Code
Overall Code
Security
0 Open issues
A
Reliability
0 Open issues
A
Maintainability
119 Open issues
A
Accepted issues
0
Valid issues that were not fixed
Coverage
0.0%
On 717 lines to cover.
Duplications
0.0%
On 2k lines.
Security Hotspots
0
A
Activity
Graph type
Issues
There isn't enough data to generate an activity graph.
May 6, 2025 at 6:08 PM
1.0
Quality Gate: Passed
First analysis: 119 Issues • 0.0% Coverage • 0.0% Duplications
See full history of analyses



8- Code Smells



8.1- Remove Useless Assignment to Local Variable "course" (AdminService.java):

Problem: You are assigning a value to the variable course but never using it afterward.

Fix: Simply remove the assignment to course if it's unnecessary, or directly return the result without storing it in the local variable.



8.2- Remove Unused "InstructorService" Private Field (InstructorServiceTest.java):

Problem: The InstructorService private field is declared but not used anywhere in the class.

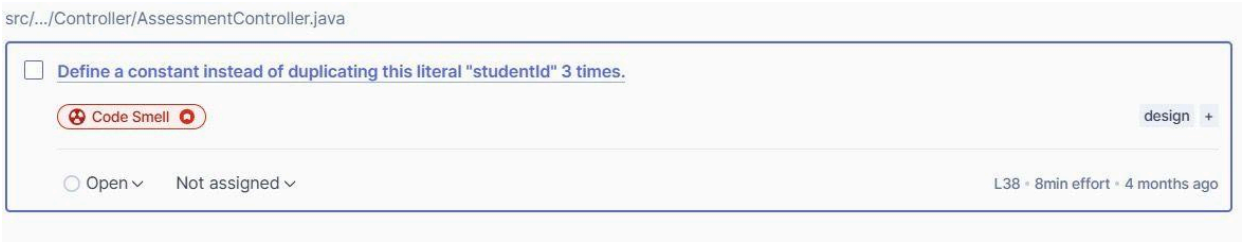
Fix: Remove the unused InstructorService private field to clean up the code.



8.3- Remove Unused Method Parameter "adminId" (Course.java):

Problem: The method has a parameter adminId that is never used within the method.

Fix: Remove the unused adminId parameter from the method definition.



8.4- Define a Constant Instead of Duplicating Literal "studentId" 3 Times (AssessmentController.java):

Problem: The literal "studentId" is used multiple times in the code, which is repetitive and error-prone. If this value needs to be updated in the future, you would have to change it in multiple places.

Fix: Define a constant for the string "studentId" and replace all its occurrences with the constant. This will make the code more maintainable and less error-prone.

9- New Features

9.1- Certificate Generator

Description: Automatically generate certificates for students upon course completion. The certificates include:

- Student name
- Course name
- Completion date
- Instructor name

This feature generates a PDF certificate that can be downloaded, providing students with an official record of their completion. The certificates can be personalized for each student based on their course completion.

Benefit: Enhances the professional aspect of your LMS, enabling students to receive a certificate for their achievements, which is useful for their career or academic progression.

Type: Enhancive

9.2- In-App Notifications for New Assignments and Quizzes

Description: Send real-time in-app notifications to students when:

1. New assignments or quizzes are posted by instructors.
2. Notifications include the content of the message and a timestamp.
3. Students can view and mark notifications as read through a notification bell in the app.

This feature ensures that students stay informed about important updates in their courses and can act quickly on new assignments or quizzes.

Benefit: Keeps students engaged and up-to-date, ensuring they do not miss important course-related updates.

Type: Enhancive

9.3- Personalized Course Recommendations

Description: Suggest personalized courses to students based on:

1. Completed courses: Suggest courses related to or building upon previously completed courses.
2. Performance trends: Recommend courses based on the student's performance (e.g., suggesting advanced courses if a student performs well).
3. Peer enrollments: Recommend courses based on the courses that similar students (peers) are enrolling in.

This recommendation system uses the student’s learning journey and behaviors to suggest relevant courses, increasing the chances that students will engage with and complete the recommended courses.

Benefit: Improves user experience by helping students discover relevant learning paths, leading to better engagement and progression through their studies.

Type: Enhancive

9.4- Student Feedback Mechanism for Courses and Instructors

Description: Enable students to provide feedback for courses and instructors after course completion. Features include:

1. Rating system (1-5 stars) for both courses and instructors.
2. A comment box for students to share their experience and suggestions.
3. Instructors can view feedback to improve teaching methods and course structure.

Benefit: Allows instructors to continuously improve their courses based on student feedback, while also providing students with a voice in their learning experience. Admins can monitor and improve overall course quality.

Type: Enhancive

9.5- Advanced Search and Filtering for Courses

Description: Provide students with an advanced search and filtering system to help them easily find relevant courses. Features include:

1. Search bar to search by keywords, course titles, or descriptions.
2. Filters to refine results by:
3. Category (e.g., "Programming", "Math") and instructor.
4. This feature enhances course discoverability and ensures that students can quickly find courses that align with their learning goals and preferences.

Benefit: Improves user experience by making it easy for students to navigate the course catalog and find courses that best suit their needs.

Type: Enhancive

9.6- Gamification: Badges

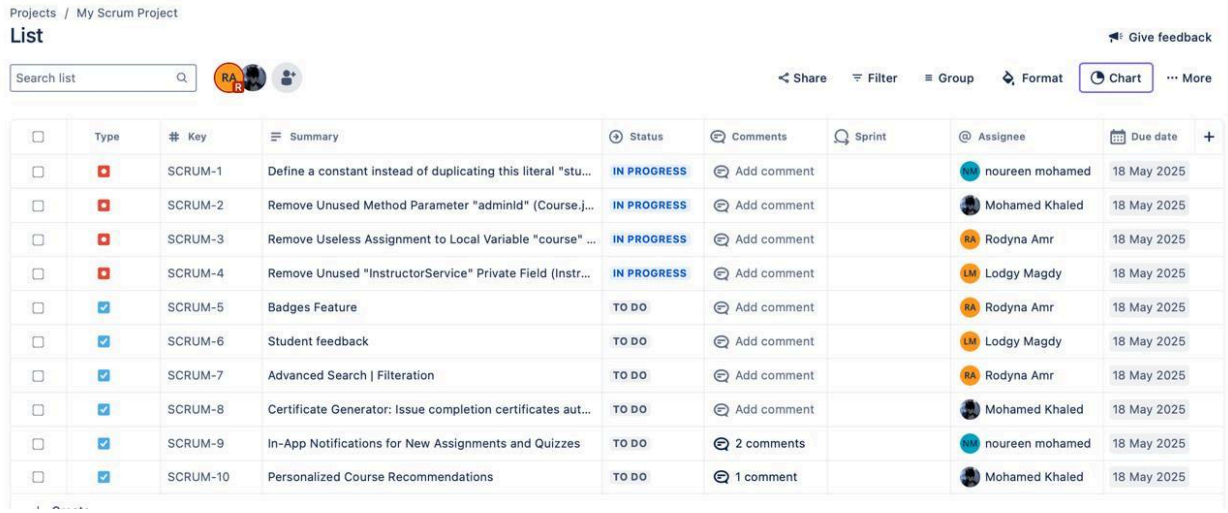
Description: Introduce a gamification layer by awarding badges to students for achieving certain milestones in their learning journey. Examples include:

1. Completion Badges: Awarded when students complete a course or specific assignments.
2. Achievement Badges: Awarded for specific actions, such as scoring perfectly on a quiz, attending all lessons, or achieving a high grade in an assignment.
3. Badge System:
   -Badges are stored as part of the student's profile
   -Badges can be awarded automatically when certain criteria are met (e.g., completing a course, achieving high scores).
   -Each badge includes a title, description, criteria (e.g., "Complete all lessons").

Benefit: Adds motivation and engagement by rewarding students for their progress and achievements. It makes the learning process more interactive and competitive, encouraging students to achieve more.

Type: Enhancive

10- Jira



Type	Key	Summary	Status	Comments	Sprint	Assignee	Due date
	SCRUM-1	Define a constant instead of duplicating this literal "stu...	IN PROGRESS	Add comment		noureen mohamed	18 May 2025
	SCRUM-2	Remove Unused Method Parameter "adminId" (Course.j...	IN PROGRESS	Add comment		Mohamed Khaled	18 May 2025
	SCRUM-3	Remove Useless Assignment to Local Variable "course" ...	IN PROGRESS	Add comment		Rodyna Amr	18 May 2025
	SCRUM-4	Remove Unused "InstructorService" Private Field (Instr...	IN PROGRESS	Add comment		Lodgy Magdy	18 May 2025
	SCRUM-5	Badges Feature	TO DO	Add comment		Rodyna Amr	18 May 2025
	SCRUM-6	Student feedback	TO DO	Add comment		Lodgy Magdy	18 May 2025
	SCRUM-7	Advanced Search | Filteration	TO DO	Add comment		Rodyna Amr	18 May 2025
	SCRUM-8	Certificate Generator: Issue completion certificates aut...	TO DO	Add comment		Mohamed Khaled	18 May 2025
	SCRUM-9	In-App Notifications for New Assignments and Quizzes	TO DO	2 comments		noureen mohamed	18 May 2025
	SCRUM-10	Personalized Course Recommendations	TO DO	1 comment		Mohamed Khaled	18 May 2025

[Github]
[Jira]

Lodgy Magdy 20226080
Rodyna Amr 20227012
Mohamed Khaled 20225056
Noureen Mohamed 20226114